

WASTEWISE-ML · DESIGN DECISIONS REGISTER

WasteWise-ml design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/WasteWise-ml		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the build plan; open items from WML-DDR-001 to WML-DDR-003 and the review note

WasteWise-ml design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction of the scanning rig: clamped base board, square tube post and arm with corner plates, foot angles, folded phone tray, cable route, mat placed by the board	Accept; change any item	Accept: each change keeps what the rig does and nothing changes the pitch or the safety case	The whole stand (build plan sections 3.1 to 3.6)	WML-DDR-003, P1 to P6
2	How the rig suits a phone whose lens is off its centre line across the width (the mat has 5.3 mm to spare front and back)	(a) cut the arm shorter by the lens offset; (b) slot the tray tab so the tray slides; (c) a 480 mm deep mat	(a): no new part	Arm length and tray window (build plan sections 3.4 and 3.6)	WML-DDR-003, A1
3	Lighting over the mat	(a) no lamp, site light recorded with each photo; (b) diffused 12 V LED strip under the arm; (c) a lamp chosen with the partner after the first site visit	(a) for the prototype; decide with field data, since a lamp changes the training photos	None for (a); a strip and its wiring for (b)	WML-DDR-003, A2
4	Sites without a suitable bench (square back edge, about 600 mm deep or more)	(a) as designed, handheld use elsewhere; (b) a separate base plate for floor or table use	(a) for the prototype	None for (a)	WML-DDR-003, A3
5	First partner organization, city and buyers	Partner, city and buyers for co-design, the field dataset and the evaluation set	None yet	Not part of the rig build; sets the reference phone, the site light and the bench	WML-DDR-001, O1
6	The CERN-OHL-S line in <code>CONTRIBUTING.md</code> in an MIT-only repository	Remove the line; keep it	None yet	None	WML-DDR-001, O2

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The reference phone's width and thickness, the size of its camera bump and how far its lens is from the centre line across the width	They set the tray's inside width, the window and the arm's length; the model assumes 76 x 9 mm, a 30 x 30 mm bump and a centred lens	WML-DDR-003, A1
2	The phone's main camera is a 26 mm-equivalent lens, or the picture size at 440 mm is measured	The picture's fit on the mat (586 x 439 mm on 700 x 450 mm) depends on it	WML-CAL-001, section 6
3	The bench at the pilot site has a square back edge, a top 20 to 50 mm thick and room under it for a G-clamp's lower jaw	The board is clamped to that edge	WML-DDR-003, P1
4	The USB cable reaches about 0.93 m with a loop to spare and fits the phone's port	The cable runs up the stand	WML-DDR-003, P5

■ Value engineering

Value-engineering target: none set (a hypothetical control target is not defined here, because `budget_usd` is null in this software repository). Estimated cost of the constructable design: USD 309 for one station, up from USD 285 because the stand is now made and clamped to the bench (WML-DDR-003), and USD 1,348 for a two-station pilot with the dataset, evaluation set and compute. Main cost drivers and savings worth trying:

- The reference phone is the largest line (USD 130), then the seven bins (USD 70), the hazard box and platform scale (USD 25 each), the power bank and cable (USD 15) and the stand clamps and fixings (USD 14).
- The dataset and evaluation set labeling effort (USD 430 and USD 260) dominates the two-station pilot; its rate is set with the partner.
- Savings worth trying: a cheaper phone that still meets R12 (about USD 150 or less, 3 GB RAM, 2019 or later chipset), and no lamp over the mat (open decision 3, option a).

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	D1 to D8: single item on a mat; MobileNetV3-Large with TensorFlow Lite and ONNX exports; keep the composite class and the unsure and hazard outputs; per-site grade file; tap-to-scan; consent, opt-in upload, no faces and co-ownership; labeling pay set with the partner at or above the local living wage; one reference phone per station for the evaluation	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	WML-DDR-001
2026-09-25	N1: at least 400 hazard items in the evaluation set; N2: ZeroWaste excluded from training; N3: camera 440 mm above the mat; O3: kit option for MIT labels on drawing sheets (cross-repo action)	Amish: "i accept all your recommendations, go with them across all repos."	WML-DDR-002